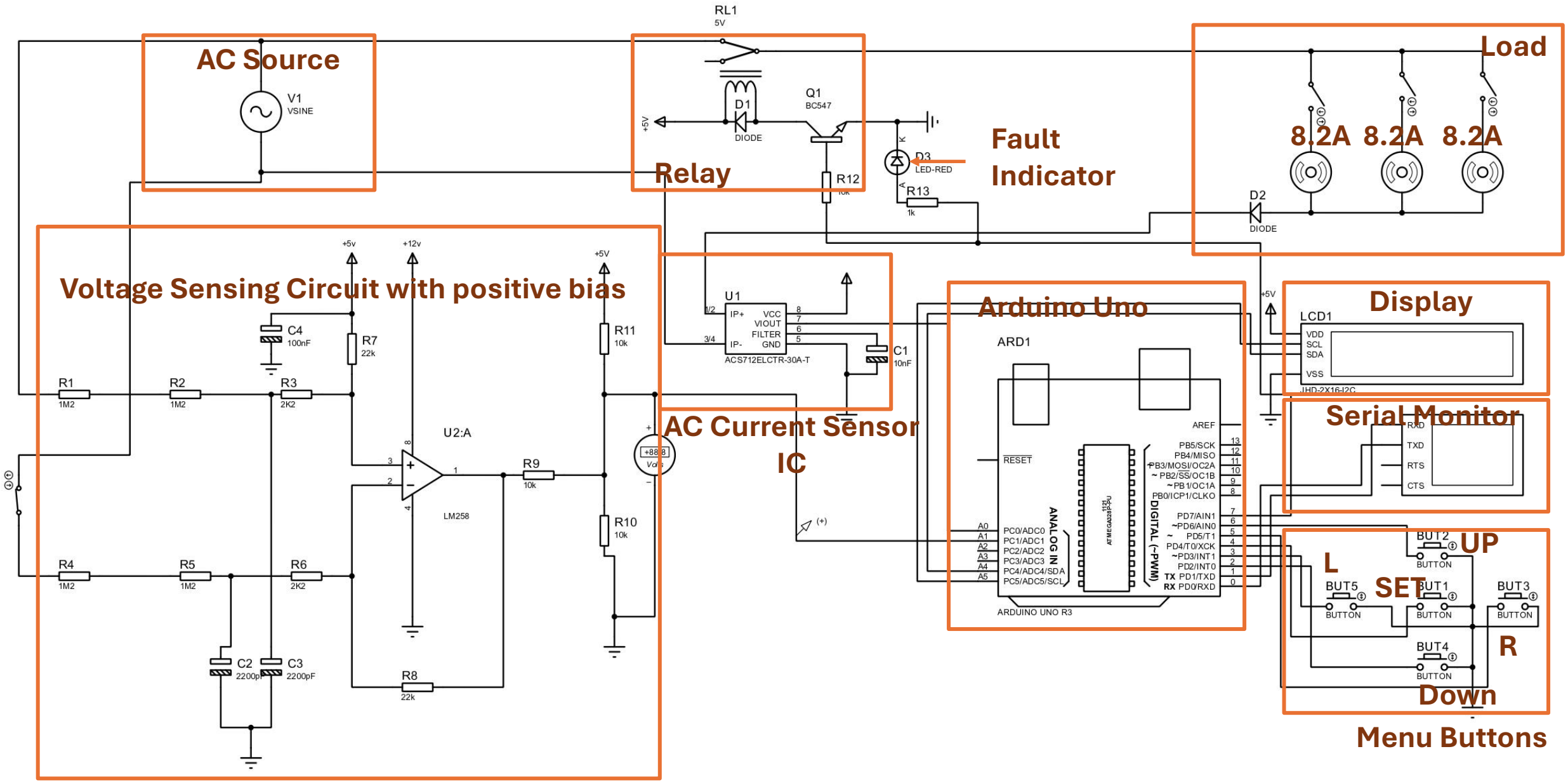

AC Source
V1
VSINE
RL1
5V
Relay
Q1
BC547
D1
DIODE
R12
D3
LED-RED
R13
1k
Fault
Indicator
Load
8.2A
8.2A
8.2A
D2
DIODE
Voltage Sensing Circuit with positive bias
+5v
+12v
+5V
C4
100nF
R7
22k
R11
10k
R1
1M2
R2
1M2
R3
2K2
U2:A
LM258
R9
10k
R10
10k
R4
1M2
R5
1M2
R6
2K2
C2
2200pF
C3
2200pF
R8
22k
U1
ACS712ELCTR-30A-T
IP+
IP-
VCC
VIOUT
FILTER
GND
C1
10nF
AC Current Sensor
IC
Arduino Uno
ARD1
ARDUINO UNO R3
RESET
AREF
ANALOG IN
DIGITAL (~PWM)
PC0/ADC0
PC1/ADC1
PC2/ADC2
PC3/ADC3
PC4/ADC4/SDA
PC5/ADC5/SCL
PB5/SCK
PB4/MISO
PB3/MOSI/OC2A
PB2/SS/OC1B
PB1/OC1A
PB0/ICP1/CLKO
PD7/AIN1
PD6/AIN0
PD5/T1
PD4/T0/XCK
PD3/INT1
PD2/INT0
TX PD1/TXD
RX PD0/RXD
Display
LCD1
VDD
SCL
SDA
VSS
Serial Monitor
RXD
TXD
RTS
CTS
BUT2
UP
L
BUT5
SET
BUT1
BUT3
R
BUT4
Down
BUTTON
Menu Buttons